

BTZ Sonic Alchemy — Testing & Validation Checklist

This document provides a systematic guide for validating the DSP and UI behavior of the BTZ plugin. Use this checklist during QA to ensure every feature sounds and acts exactly as intended.

1. Core Saturation & Drive

Test Setup: Feed a clean, dynamic drum loop or a pure sine wave (1kHz) into the plugin.

Feature	Action	What You Should Hear / See
Drive Knob	Turn from 0% to 100%	The signal should become increasingly distorted and harmonically rich. Crucially, the overall perceived volume should remain relatively constant due to the Auto-Gain system. You should hear density, not just loudness.
Tube Model	Select “Tube”, push Drive	Warm, thick saturation. On a sine wave, you should see strong 2nd and 4th order (even) harmonics in the visualizer. Transients should round off smoothly.
Tape Model	Select “Tape”, push Drive	Symmetrical clipping. You should hear odd harmonics (3rd, 5th) and a slight softening of the extreme high frequencies.
Transformer	Select “Transformer”, feed low freq	You should hear low-frequency weight and hysteresis. The sub frequencies should feel “anchored” and slightly compressed.
Transistor	Select “Transistor”, push Drive	Aggressive, biting distortion. The top end should sound fizzy and the clipping should feel harder than the Tube model.

2. Dynamics & Tone Shaping

Test Setup: Feed a full mix bus or a complex synth chord progression.

Feature	Action	What You Should Hear / See
Tone Knob	Sweep from 0% to 100%	At 0%, the signal should sound dark and muffled. At 50%, it should be neutral. At 100%, it should sound bright and airy, but harsh resonant peaks should be dynamically suppressed (not just a static EQ boost).
Punch Knob	Turn from 0% to 100% on drums	At 0%, the drum transients should be flattened by the saturation. At 100%, the initial stick attack of the snare should poke through cleanly before the saturation clamps down on the tail.
Density (Glue)	Increase Density on a mix bus	The mix should feel more cohesive. You should hear the quiet details come up and the loud peaks tuck in. The gain reduction meter should show 2-4dB of movement.
Sidechain HPF	Toggle HPF while compressing	With HPF on, a heavy kick drum should no longer cause the entire mix to “pump” or duck. The compression should react more to the snare and vocals.

3. Advanced Processing

Test Setup: Feed a wide stereo acoustic guitar or piano track.

Feature	Action	What You Should Hear / See
Multiband Split	Enable 2 bands, solo the low band	You should hear only the frequencies below the crossover point. The crossover should be completely transparent (no phase smearing) when bands are summed back together.
Mid/Side Mode	Enable M/S, push Drive on Side	The center of the image (vocals, kick, bass) should remain clean, while the wide stereo information (reverbs, wide guitars) becomes saturated and harmonically rich.
Width Slider	Sweep from 0% to 200%	At 0%, the signal should collapse to perfect mono. At 100%, it should be unchanged. At 200%, the side information should be exaggerated, making the mix sound artificially wide.
LFO Modulation	Assign LFO to Drive, set to Sine	You should hear the saturation amount rhythmically swelling in and out. The UI Drive knob should visually indicate the modulation range.

4. Utility & Metering

Test Setup: Feed any commercial reference track.

Feature	Action	What You Should Hear / See
Delta (Δ) Mode	Click the Delta button	You should hear <i>only</i> the difference between the wet and dry signals. If Drive is at 0%, you should hear silence. If Drive is high, you should hear a thin, fizzy, distorted signal (the generated harmonics).
Mix Slider	Sweep from 0% to 100%	The transition should be perfectly smooth. There should be absolutely no comb filtering or phase cancellation (hollow sound) during the sweep, proving latency compensation is correct.
True Peak Limiter	Push Master output, set Ceiling to -1dB	The output meter should never exceed -1.0 dBTP, no matter how hard you push the input. You should not hear digital clipping clicks.
Correlation Meter	Feed a mono signal, then a wide signal	Mono signal: Meter reads +1. Wide signal: Meter hovers around 0 to +0.5. Out-of-phase signal: Meter drops below 0.
Bypass	Toggle Bypass on and off	The transition should be click-free (a fast 64-sample crossfade occurs internally).

5. UI & Interaction

Feature	Action	What You Should Hear / See
Mode Switching	Click Simple, Standard, Advanced tabs	The UI should instantly resize and reveal/hide controls. Audio should not glitch or drop out during the switch.
Knob Interaction	Drag a knob up/down	The movement should feel smooth. Holding Shift while dragging should enable fine-tuning (slower movement). Double-clicking should reset to default.
Undo/Redo	Change a parameter, click Undo	The parameter should revert to its previous state. Audio should reflect the change immediately.
A/B Comparison	Set state A, click B, change state, click A	You should be able to toggle between two completely different plugin states instantly without audio dropouts.